

GVA Downloader

v2.0

FFmpeg Installation Guide

Complete setup guide for Windows, Android (Termux), Linux, and macOS — configure FFmpeg so every media-processing feature of GVA Downloader v2.0 works correctly.

Fast • Portable • Cross-Platform

ENGINE

yt-dlp

INTERFACE

Rich Terminal

AUTHOR

Jeevanantham K

Table of Contents

GVA Downloader v2.0 — FFmpeg Installation Guide

1	What Is FFmpeg?	03
2	Before Installing FFmpeg	04
3	Windows — Recommended Installation	05
	Step 1 — Open the Official FFmpeg Download Page	
	Step 2 — Download a Windows Build	
	Step 3 — Extract FFmpeg	
	Step 4 — Add FFmpeg to Windows PATH	
	Step 5 — Restart the Terminal	
	Step 6 — Verify FFmpeg	
4	Start GVA Downloader	09
5	Troubleshooting — Windows	10
6	Android — Termux	11
7	Linux — Ubuntu / Debian	12
8	Fedora	13
9	Arch Linux	14
10	macOS	15
11	How to Check FFmpeg	16
12	What Happens Without FFmpeg?	17
13	Optional — Portable FFmpeg	18
14	Quick Installation Checklist	19
15	Quick Command Cheat Sheet	20
16	Final Support Section	21

01 What Is FFmpeg?

GVA Downloader v2.0 uses **yt-dlp** as its download engine to fetch video and audio from supported sites, and it uses **FFmpeg** as its media-processing engine. yt-dlp handles the downloading; FFmpeg handles converting, merging, and packaging the media afterward.

FFmpeg is required whenever GVA Downloader needs to do something more than save a single file exactly as it was downloaded. Specifically, FFmpeg is required for:

- Merging separate video and audio streams into a single playable file
- Converting audio to **MP3**
- Converting audio to **M4A**
- Converting audio to **AAC**
- Converting audio to **FLAC**
- Converting audio to **OGG**
- Converting audio to **WAV**
- Embedding thumbnails and cover art into finished files
- Some metadata operations (tagging, chapters, and similar post-processing)
- Other general media post-processing tasks

GOOD TO KNOW

Some simple, direct downloads may still work even without FFmpeg installed — for example, when a site already serves a single combined file. However, any feature that involves converting, merging, or embedding media will fail without FFmpeg.

GVA Downloader v2.0 automatically checks whether FFmpeg is available every time it starts up. If FFmpeg cannot be found, GVA Downloader will display a warning so you know that some features may not work correctly until it is installed.

02 Before Installing FFmpeg

Before you install FFmpeg, it helps to understand exactly what GVA Downloader v2.0 needs in order to run correctly.

Required Components

- **GVA Downloader v2.0** — the application itself
- **Python** — the runtime GVA Downloader is built on
- **yt-dlp** — the download engine
- **Rich** — powers the terminal interface
- **FFmpeg** — the media-processing engine (this guide)

FFmpeg is **not** a Python package and is **not** installed automatically through pip. It is a separate, external program that must be installed on your operating system and made available on your system's **PATH** so that GVA Downloader (and any other program) can find and run it.

IMPORTANT

Do not download FFmpeg from random websites. Use the official FFmpeg website, or a Windows build provider linked directly from the official FFmpeg download page.

Official FFmpeg website: <https://ffmpeg.org/download.html>

03 FFmpeg Installation on Windows 10/11

Windows is the primary platform for GVA Downloader v2.0, so this section walks through every step in detail — including for readers who have never edited a PATH variable before.

Step 1 — Open the Official FFmpeg Download Page

Open your browser and go to the official FFmpeg download page:

```
https://ffmpeg.org/download.html
```

The official FFmpeg site links out to trusted Windows executable builds rather than hosting the Windows `.exe` files itself. From the Windows section of that page, you will find links to reputable build providers such as **gyan.dev** and **BtbN**. These are the build providers referenced directly from the official FFmpeg website.

TIP

If you are unsure which build to pick, a "release" full build (not "essentials", not a nightly/git build) from one of these official-linked providers is the safest choice for most users.

Step 2 — Download a Windows Build

Download a current **64-bit Windows build**. The file you download will normally be a **ZIP archive**. Its internal folder structure will look similar to this:

```
ffmpeg-...  
├─ bin  
│   ├── ffmpeg.exe  
│   ├── ffprobe.exe  
│   └─ ffplay.exe
```

DON'T WORRY ABOUT THE EXACT NAME

Do not worry if the exact FFmpeg filename or version differs from the example above. FFmpeg versions change over time, and the filename will normally include a version number and build date.

Step 3 — Extract FFmpeg

1. Right-click the downloaded ZIP file.

2. Select "**Extract All...**"
3. Extract it somewhere simple and easy to remember.

A recommended, simple extraction location is:

```
C:\ffmpeg
```

After extraction, your folder should look like this:

```
C:\ffmpeg\  
└─ bin\  
    ├── ffmpeg.exe  
    ├── ffprobe.exe  
    └─ ffplay.exe
```

IMPORTANT — PATH MUST POINT TO THE BIN FOLDER

The PATH entry must point to `C:\ffmpeg\bin` — the folder that directly contains `ffmpeg.exe`.

NOT `C:\ffmpeg`

NOT `C:\ffmpeg\bin\ffmpeg.exe`

Step 4 — Add FFmpeg to Windows PATH

This is the most important step. Follow it carefully — even if you have never edited an environment variable before.

1. Press the **Windows key** on your keyboard.
2. Type "**Environment Variables**" and look at the search results.
3. Open "**Edit the system environment variables.**"
4. In the System Properties window that opens, click the "**Advanced**" tab.
5. Click the "**Environment Variables...**" button near the bottom.

You will now see two lists: **User variables** and **System variables**. Both work — pick whichever fits your situation:

Option	When to use it
User variables	Recommended for a personal computer used by one person.
System variables	Makes FFmpeg available to every user account on this computer.

In whichever list you chose:

1. Find and select the variable named "**Path.**"
2. Click "**Edit.**"
3. Click "**New.**"
4. Type in the following folder path exactly:

```
C:\ffmpeg\bin
```

1. Click **OK** to close the Edit Environment Variable window.
2. Click **OK** again to close the Environment Variables window.
3. Close all remaining settings windows.

Step 5 — Restart the Terminal

Any Command Prompt, PowerShell, or Git Bash window that was already open **will not** see the new PATH value. You must open a fresh terminal window for the change to take effect.

1. Close every open Command Prompt / PowerShell / Git Bash window.
2. Open a brand new terminal window.

Step 6 — Verify FFmpeg

In your new terminal window, run:

```
ffmpeg -version
```

A successful result will print FFmpeg's version number, build configuration, and library versions — several lines of text starting with something like `ffmpeg version ...`. If instead you see an error such as *"not recognized as an internal or external command,"* the PATH was not set correctly — see the troubleshooting section below.

Next, confirm exactly where Windows is finding FFmpeg:

```
where ffmpeg
```

The expected result should look similar to:

```
C:\ffmpeg\bin\ffmpeg.exe
```

Finally, check that `ffprobe` (used for reading media information) is also available:

```
ffprobe -version
```

NOTE

The exact version numbers you see will differ from the examples in this guide — that is expected, since FFmpeg is updated regularly.

04 Start GVA Downloader

Once FFmpeg is verified, navigate to the folder containing GVA Downloader v2.0:

```
cd "C:\Path\To\GVA Downloader"
```

Then start it:

```
python gva_downloader.py
```

GVA Downloader v2.0 performs a startup dependency check every time it launches. If your PATH is configured correctly, you should no longer see the FFmpeg-missing warning, and all conversion and merging features should now work as expected.

05 Troubleshooting — Windows

"'ffmpeg' is not recognized as an internal or external command"

- Confirm that `ffmpeg.exe` actually exists at `C:\ffmpeg\bin\ffmpeg.exe`.
- Verify your PATH variable contains exactly: `C:\ffmpeg\bin`
- Close and reopen your terminal — PATH changes never apply to already-open windows.
- Run `where ffmpeg` to confirm Windows can locate it.

"FFmpeg works in CMD but GVA Downloader says it is missing"

1. Close GVA Downloader.
2. Close the terminal window.
3. Open a brand new terminal window.
4. Run `where ffmpeg`
5. Run `ffmpeg -version`
6. Start GVA Downloader again.

"Wrong PATH was added"

WRONG

`C:\ffmpeg`

WRONG

`C:\ffmpeg\bin\ffmpeg.exe`

CORRECT

`C:\ffmpeg\bin`

"Downloaded FFmpeg but cannot find ffmpeg.exe"

This usually means the ZIP extracted into an extra nested folder. Look for a structure similar to:

Possible:
`C:\ffmpeg\ffmpeg-...\bin\ffmpeg.exe`

Your PATH entry must point to the folder that **directly contains** `ffmpeg.exe` — not a parent folder above it.

06 Android — Termux

Android users running GVA Downloader v2.0 through Termux can install FFmpeg with just a couple of commands.

```
pkg update -y  
pkg install ffmpeg -y
```

Verify the installation:

```
ffmpeg -version
```

Then start GVA Downloader:

```
python gva_downloader.py
```

TIP

The GVA Downloader Termux installer can also install FFmpeg automatically as part of its setup process.

07 Linux — Ubuntu / Debian

```
sudo apt update  
sudo apt install ffmpeg -y
```

Verify the installation:

```
ffmpeg -version
```

Then start GVA Downloader:

```
python3 gva_downloader.py
```

08 Fedora

```
sudo dnf install ffmpeg
```

Verify the installation:

```
ffmpeg -version
```

09 Arch Linux

```
sudo pacman -S ffmpeg
```

Verify the installation:

```
ffmpeg -version
```

10 macOS

On macOS, the simplest way to install FFmpeg is with **Homebrew**.

```
brew install ffmpeg
```

Verify the installation:

```
ffmpeg -version
```

Then start GVA Downloader:

```
python3 gva_downloader.py
```

11 How to Check FFmpeg

Use the table below to confirm FFmpeg is correctly available on your platform:

Platform	Command
Windows	<code>where ffmpeg</code>
Linux / macOS / Termux	<code>which ffmpeg</code>
All platforms	<code>ffmpeg -version</code>

If the command returns FFmpeg's file location and/or version information, FFmpeg is normally available through your system PATH and GVA Downloader v2.0 should be able to detect it automatically.

12 What Happens Without FFmpeg?

yt-dlp handles the downloading, while FFmpeg handles the media processing. The table below shows which GVA Downloader v2.0 features depend on FFmpeg:

GVA Downloader Feature	FFmpeg Required?
Some direct downloads	Sometimes no
Separate video/audio merging	YES
MP3 conversion	YES
M4A conversion	YES
AAC conversion	YES
FLAC conversion	YES
OGG conversion	YES
WAV conversion	YES
Thumbnail embedding	YES
Some metadata / post-processing	YES

13 Optional — Portable FFmpeg

FFmpeg could technically be placed directly inside the GVA Downloader project directory, for example:

```
GVA Downloader/  
├─ gva_downloader.py  
├─ ffmpeg/  
│   └─ bin/  
│       ├── ffmpeg.exe  
│       ├── ffprobe.exe  
│       └─ ffplay.exe  
├─ downloads/  
├─ config/  
├─ history/  
├─ logs/  
├─ cache/  
└─ temp/
```

VERY IMPORTANT

The current GVA Downloader v2.0 script expects FFmpeg to be available as a system executable/PATH dependency. Bundled FFmpeg is **not** currently an automatic feature of the v2.0 script. Placing FFmpeg inside the project folder as shown above will not, by itself, make GVA Downloader v2.0 detect it — FFmpeg still needs to be installed on your system PATH using the steps in this guide.

14 Quick Installation Checklist

Print this page, or keep it open, as you work through your installation.

Windows

- ☐ Download FFmpeg from the official source
- ☐ Extract FFmpeg
- ☐ Locate the bin folder
- ☐ Confirm ffmpeg.exe exists
- ☐ Add C:\ffmpeg\bin to PATH
- ☐ Close and reopen terminal
- ☐ Run ffmpeg -version
- ☐ Run where ffmpeg
- ☐ Start GVA Downloader v2.0
- ☐ Test a download

Linux

- ☐ Install FFmpeg using package manager
- ☐ Run ffmpeg -version
- ☐ Start GVA Downloader

macOS

- ☐ brew install ffmpeg
- ☐ ffmpeg -version
- ☐ Start GVA Downloader

Termux

- ☐ pkg update -y
- ☐ pkg install ffmpeg -y
- ☐ ffmpeg -version
- ☐ Start GVA Downloader

15 Quick Command Cheat Sheet

Windows

```
ffmpeg -version  
where ffmpeg  
ffprobe -version  
python gva_downloader.py
```

Termux

```
pkg update -y  
pkg install ffmpeg -y  
ffmpeg -version  
python gva_downloader.py
```

Ubuntu / Debian

```
sudo apt update  
sudo apt install ffmpeg -y  
ffmpeg -version  
python3 gva_downloader.py
```

Fedora

```
sudo dnf install ffmpeg  
ffmpeg -version
```

Arch

```
sudo pacman -S ffmpeg  
ffmpeg -version
```

macOS

```
brew install ffmpeg  
ffmpeg -version  
python3 gva_downloader.py
```

16 Final Support Section

If FFmpeg is installed but GVA Downloader still reports it as missing, first verify that `ffmpeg -version` works in a **newly opened** terminal. Then check that the FFmpeg `bin` directory is correctly added to your PATH.

Reminder: GVA Downloader v2.0 uses yt-dlp for downloading and FFmpeg for media processing.

GVA Downloader v2.0 — FFmpeg Installation Guide • Author: Jeevanantham K • Engine: yt-dlp • Interface: Rich Terminal